

```
``html

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>VoltraBloom — Hybrid Energy Harvesting System</title>

<script src="https://cdn.tailwindcss.com"></script>

<script src="https://unpkg.com/lucide@latest"></script>

<script src="https://cdn.jsdelivr.net/npm/three@0.128.0/examples/js/r128/three.min.js"></script>

<script
src="https://cdn.jsdelivr.net/npm/three@0.128.0/examples/js/controls/OrbitControls.js"></script>

<script src="https://cdn.jsdelivr.net/npm/gsap@3.12.5/gsap.min.js"></script>

<script src="https://cdn.jsdelivr.net/npm/gsap@3.12.5/ScrollTrigger.min.js"></script>

<link
href="https://fonts.googleapis.com/css2?family=DM+Serif+Display&family=Inter:wght@300;400;500;600;700&display=swap" rel="stylesheet">

<script>

tailwind.config={theme:{extend:{fontFamily:{serif:['DM Serif Display','serif'],sans:['Inter','sans-serif']}}}}

</script>

<style>

*{margin:0;padding:0;box-sizing:border-box}html{scroll-behavior:smooth}

body{font-family:'Inter',sans-serif;background:#FAF6EE;color:#261E14;overflow-x:hidden;-webkit-font-smoothing:antialiased}

::-webkit-scrollbar{width:5px}::-webkit-scrollbar-track{background:#FAF6EE}::-webkit-scrollbar-thumb{background:#c4b9a8;border-radius:3px}

.grid-matrix{position:fixed;inset:0;pointer-events:none;z-index:1;background-image:linear-gradient(rgba(38,30,20,.04) 1px,transparent 1px),linear-gradient(90deg,rgba(38,30,20,.04) 1px,transparent 1px);background-size:80px 80px}

#c3d{position:fixed;inset:0;z-index:0}

.content-layer{position:relative;z-index:10;pointer-events:none}

.content-layer *{pointer-events:none}

.interactive-el{pointer-events:auto !important}
```

```

.step-block{min-height:100vh;display:flex;align-
items:center;opacity:0;transform:translateY(30px);transition:opacity .7s ease,transform .7s ease}

.step-block.visible{opacity:1;transform:translateY(0)}

.telemetry-val{font-variant-numeric:tabular-nums}

@keyframes pulse-glow{0%,100%{box-shadow:0 0 4px rgba(77,124,15,.3)}50%{box-shadow:0 0 12px
rgba(77,124,15,.6)}}

.glow-indicator{animation:pulse-glow 2s ease-in-out infinite}

</style>

</head>

<body>

<div class="grid-matrix"></div>

<canvas id="c3d"></canvas>

<div class="content-layer">

<nav class="fixed top-0 left-0 right-0 z-50 flex items-center justify-between px-6 md:px-10 py-4"
style="background:rgba(250,246,238,.85);backdrop-filter:blur(12px);border-bottom:1px solid
rgba(38,30,20,.06)">

<div class="flex items-center gap-2.5"><div class="w-7 h-7 rounded-md flex items-center justify-
center" style="background:#E2A03F"><i data-lucide="sun" class="w-4 h-4 text-
white"></i></div><span class="text-sm font-bold tracking-tight"
style="color:#261E14">VoltraBloom</span></div>

<div class="hidden md:flex items-center gap-6 text-xs tracking-wider uppercase"
style="color:#8a7e6e"><a href="#step-1" class="interactive-el hover:text-[#261E14] transition-
colors">System</a><a href="#step-3" class="interactive-el hover:text-[#261E14] transition-
colors">Controller</a><a href="#step-5" class="interactive-el hover:text-[#261E14] transition-
colors">Output</a></div>

</nav>

<section class="min-h-screen flex flex-col items-center justify-center text-center px-6 pt-20">

<div class="mb-4 inline-flex items-center gap-2 px-3 py-1.5 rounded-full text-[10px] tracking-[.15em]
uppercase" style="background:rgba(38,30,20,.04);color:#8a7e6e;border:1px solid
rgba(38,30,20,.06)"><span class="w-1.5 h-1.5 rounded-full bg-[#E2A03F]"></span>Hybrid Energy
Harvesting System</div>

<h1 class="text-5xl md:text-7xl font-serif tracking-tight leading-[.95] mb-5"
style="color:#261E14">VoltraBloom</h1>

```

<p class="text-sm md:text-base max-w-lg leading-relaxed mb-10" style="color:#8a7e6e">Solar, wind, and bioelectric convergence in a single deployable unit. Clean reserve power engineered for critical building infrastructure.</p>

<div class="flex items-center gap-4">

<i data-lucide="eye" class="w-3.5 h-3.5"></i>View Project Poster

<i data-lucide="arrow-up-right" class="w-3.5 h-3.5"></i>Live Telemetry Data

</div>

<div class="absolute bottom-10 left-1/2 -translate-x-1/2 flex flex-col items-center gap-2 opacity-40">Scroll to explore<i data-lucide="chevron-down" class="w-4 h-4"></i></div>

</section>

<div id="step-1" class="step-block px-6 md:px-10 lg:px-16"><div class="max-w-xl"><div class="text-[10px] tracking-[.2em] uppercase mb-3" style="color:#c4b9a8">Step 01</div><h2 class="text-3xl md:text-4xl font-serif tracking-tight leading-[.95] mb-4" style="color:#261E14">Hybrid Energy Harvesting</h2><p class="text-sm leading-relaxed" style="color:#6b5f50">VoltraBloom initiates energy capture by combining independent natural forces simultaneously. The upper monocrystalline solar petals deploy to track peak sun hours, while the vertical-axis wind turbine captures kinetic air currents. Concurrently, the integrated Soil Microbial Fuel Cells (SMFC) draw steady milliwatt-level current from biochemical processes in the soil substrate.</p></div></div>

<div id="step-2" class="step-block px-6 md:px-10 lg:px-16"><div class="max-w-xl"><div class="text-[10px] tracking-[.2em] uppercase mb-3" style="color:#c4b9a8">Step 02</div><h2 class="text-3xl md:text-4xl font-serif tracking-tight leading-[.95] mb-4" style="color:#261E14">Power Conditioning Circuit</h2><p class="text-sm leading-relaxed" style="color:#6b5f50">Raw electrical currents generated from the solar and wind nodes vary constantly due to shifting environmental conditions. The power lines feed directly into a high-efficiency 48V/80A MPPT Charge Controller and an AC Rectifier inside the base. This architecture optimizes the voltage curves in real-time, ensuring maximum power point tracking across all three energy harvesters without oscillation loss.</p></div></div>

<div id="step-3" class="step-block px-6 md:px-10 lg:px-16"><div class="max-w-xl"><div class="text-[10px] tracking-[.2em] uppercase mb-3" style="color:#c4b9a8">Step 03</div><h2 class="text-3xl

md:text-4xl font-serif tracking-tight leading-[.95] mb-4" style="color:#261E14">Embedded HEMS Controller</h2><p class="text-sm leading-relaxed" style="color:#6b5f50">The Hybrid Energy Management System (HEMS), driven by an ESP32 microcontroller, serves as the central neural network. It continuously processes live diagnostic telemetry from digital current, voltage, and temperature sensors, transmitting data instantly via IoT protocols to our remote database for predictive maintenance and performance analytics.</p></div></div>

<div id="step-4" class="step-block px-6 md:px-10 lg:px-16"><div class="max-w-xl"><div class="text-[10px] tracking-[.2em] uppercase mb-3" style="color:#c4b9a8">Step 04</div><h2 class="text-3xl md:text-4xl font-serif tracking-tight leading-[.95] mb-4" style="color:#261E14">High-Density Battery Cells</h2><p class="text-sm leading-relaxed" style="color:#6b5f50">Conditioned energy is routed safely into a heavy-duty 48V 200Ah LiFePO4 battery bank housed inside the protective inner chassis. Operating at an optimized 80% Depth of Discharge (DoD), this 9.6 kWh configuration stores ample clean reserve power, guaranteeing a reliable 7 to 8 hours of continuous backup autonomy during grid interruptions.</p></div></div>

<div id="step-5" class="step-block px-6 md:px-10 lg:px-16"><div class="max-w-xl"><div class="text-[10px] tracking-[.2em] uppercase mb-3" style="color:#c4b9a8">Step 05</div><h2 class="text-3xl md:text-4xl font-serif tracking-tight leading-[.95] mb-4" style="color:#261E14">Building AC Distribution</h2><p class="text-sm leading-relaxed" style="color:#6b5f50">When main grid power interruptions trigger the system, a 5 kW pure sine wave inverter instantly activates. It transforms the stored 48V DC energy into standard 220V AC electricity. This current flows seamlessly through heavy-duty breakers directly into the building main distribution panel to run critical load lines without perceptible switching delay.</p></div></div>

<div class="fixed bottom-6 left-6 z-40 w-56 rounded-lg p-4 shadow-lg" style="background:rgba(250,246,238,.95);border:1px solid rgba(77,124,15,.12);backdrop-filter:blur(8px)">

<div class="flex items-center gap-2 mb-3"><div class="w-2 h-2 rounded-full bg-[#4D7C0F] glow-indicator"></div>SMFC Diagnostics</div>

<div class="space-y-2 text-[10px]" style="color:#6b5f50">

<div class="flex justify-between">Voltage4.82V</div>

<div class="flex justify-between">Current12.3 mA</div>

<div class="flex justify-between">Power59.0 mW</div>

<div class="flex justify-between">Moisture42%</div>

```
<div class="flex justify-between"><span>Temp</span><span class="telemetry-val font-medium" style="color:#261E14">23.4 C</span></div>
```

```
<div class="flex justify-between items-center"><span>Status</span><span class="inline-flex items-center gap-1 font-medium" style="color:#4D7C0F"><span class="w-1.5 h-1.5 rounded-full bg-[#4D7C0F]"></span>ACTIVE</span></div>
```

```
<div class="w-full h-1 rounded-full mt-1" style="background:rgba(77,124,15,.1)"><div class="h-full rounded-full bg-[#4D7C0F]" style="width:67%;transition:width 1s"></div></div>
```

```
</div>
```

```
</div>
```

```
<footer class="relative z-10 py-10 text-center border-t" style="border-color:rgba(38,30,20,.06)">
```

```
<p class="text-[10px] tracking-wider" style="color:#c4b9a8">&copy; 2025 VoltraBloom &mdash; Hybrid Energy Harvesting System. All specifications subject to field conditions.</p>
```

```
</footer>
```

```
</div>
```

```
<script>
```

```
function initIcons(){if(typeof lucide!=='undefined'){lucide.createIcons()}else{setTimeout(initIcons,100)}}initIcons();
```

```
window.addEventListener('DOMContentLoaded',function(){
```

```
var camera=null,scene=null,controls=null,scrollSteps=[];
```

```
var canvas=document.getElementById('c3d');
```

```
if(!canvas){return}
```

```
var renderer=new THREE.WebGLRenderer({canvas:canvas,antialias:true,alpha:true});
```

```
renderer.setPixelRatio(Math.min(window.devicePixelRatio,2));
```

```
renderer.setSize(window.innerWidth>window.innerHeight);
```

```
renderer.toneMapping=THREE.ACESFilmicToneMapping;
```

```
renderer.toneMappingExposure=1.2;
```

```
renderer.outputEncoding=THREE.sRGBEncoding;
```

```
renderer.setClearColor(0xFAF6EE,1);
```

```
scene=new THREE.Scene();
scene.fog=new THREE.FogExp2(0xFAF6EE,0.015);

camera=new THREE.PerspectiveCamera(36,window.innerWidth/window.innerHeight,0.1,300);
camera.position.set(-5,35,45);

controls=new THREE.OrbitControls(camera,canvas);
controls.enableDamping=true;
controls.dampingFactor=0.05;
controls.enablePan=false;
controls.minDistance=15;
controls.maxDistance=120;
controls.target.set(0,5,0);

var ambLight=new THREE.AmbientLight(0x90A080,0.5);
scene.add(ambLight);
var sunLight=new THREE.DirectionalLight(0xFFFF5E6,1.8);
sunLight.position.set(10,20,10);
sunLight.castShadow=true;
sunLight.shadow.mapSize.set(1024,1024);
sunLight.shadow.camera.near=0.5;
sunLight.shadow.camera.far=100;
sunLight.shadow.camera.left=-30;
sunLight.shadow.camera.right=30;
sunLight.shadow.camera.top=30;
sunLight.shadow.camera.bottom=-30;
scene.add(sunLight);
var fillLight=new THREE.DirectionalLight(0xC0B090,0.3);
fillLight.position.set(-8,5,-5);
scene.add(fillLight);
```

```
var glassMat=new
THREE.MeshPhysicalMaterial({color:0xfffff,metalness:0,roughness:0.1,transmission:0.9,clearcoat:1.0
,clearcoatRoughness:0.1,ior:1.5,transparent:true,side:THREE.DoubleSide,thickness:0.5});

var bodyMat=new
THREE.MeshPhysicalMaterial({color:0xD8D0C4,metalness:0.05,roughness:0.35,clearcoat:0.3,clearcoa
tRoughness:0.4});

var darkMat=new THREE.MeshStandardMaterial({color:0x3A3428,metalness:0.1,roughness:0.6});

var coreMat=new
THREE.MeshStandardMaterial({color:0x4D7C0F,emissive:0x4D7C0F,emissiveIntensity:0.4,metalness:
0.2,roughness:0.4});

var accentMat=new THREE.MeshStandardMaterial({color:0xE2A03F,metalness:0.3,roughness:0.4});

var wireOrangeMat=new THREE.LineBasicMaterial({color:0xE2A03F,linewidth:1});

var wireGreenMat=new THREE.LineBasicMaterial({color:0x4D7C0F,linewidth:1});


var mainGroup=new THREE.Group();

scene.add(mainGroup);


var box1Geo=new THREE.BoxGeometry(22,15,22);
var box1=new THREE.Mesh(box1Geo,glassMat);
box1.position.set(0,7.5,0);
box1.castShadow=true;
box1.receiveShadow=true;
mainGroup.add(box1);


var lidPivot=new THREE.Group();
lidPivot.position.set(0,15,-11);
var lidGeo=new THREE.BoxGeometry(21.8,0.4,21.8);
var lidMesh=new THREE.Mesh(lidGeo,bodyMat);
lidMesh.position.set(0,0.2,11);
lidPivot.add(lidMesh);
lidPivot.rotation.x=-Math.PI*0.12;
mainGroup.add(lidPivot);
```

```
var slotGeo=new THREE.BoxGeometry(9.6,3.9,0.15);  
var slotMesh=new THREE.Mesh(slotGeo,darkMat);  
slotMesh.position.set(0,5,11.05);  
mainGroup.add(slotMesh);
```

```
var box2Geo=new THREE.BoxGeometry(22,5,22);  
var box2=new THREE.Mesh(box2Geo,bodyMat);  
box2.position.set(22,2.5,0);  
box2.castShadow=true;  
box2.receiveShadow=true;  
mainGroup.add(box2);
```

```
var coreGeo=new THREE.BoxGeometry(8.5,12,8.5);  
var core=new THREE.Mesh(coreGeo,coreMat);  
core.position.set(-4,6,4);  
mainGroup.add(core);
```

```
var coreInnerGeo=new THREE.BoxGeometry(6,9,6);  
var coreInnerMat=new  
THREE.MeshStandardMaterial({color:0x2A4A08,emissive:0x4D7C0F,emissiveIntensity:0.8});  
var coreInner=new THREE.Mesh(coreInnerGeo,coreInnerMat);  
coreInner.position.set(-4,6,4);  
mainGroup.add(coreInner);
```

```
var portGeo=new THREE.TorusGeometry(0.9,0.06,8,24);  
var port1=new THREE.Mesh(portGeo,accentMat);  
port1.position.set(11,0,0);  
port1.rotation.y=Math.PI/2;  
mainGroup.add(port1);
```

```
var port2=new THREE.Mesh(portGeo.clone(),accentMat);
```



```
port2.position.set(0,0,5);
```

```
mainGroup.add(port2);
```

```
var greenPts=[new THREE.Vector3(-4,6,8.25),new THREE.Vector3(-4,6,10),new THREE.Vector3(-4,5,10),new THREE.Vector3(0,5,10),new THREE.Vector3(0,5,11)];
```

```
var greenGeo=new THREE.BufferGeometry().setFromPoints(greenPts);
```

```
mainGroup.add(new THREE.Line(greenGeo,wireGreenMat));
```

```
var orangePts=[new THREE.Vector3(0,5,11),new THREE.Vector3(5,5,11),new THREE.Vector3(5,2.5,11),new THREE.Vector3(5,2.5,0),new THREE.Vector3(11,2.5,0)];
```

```
var orangeGeo=new THREE.BufferGeometry().setFromPoints(orangePts);
```

```
mainGroup.add(new THREE.Line(orangeGeo,wireOrangeMat));
```

```
var greenPts2=[new THREE.Vector3(-4,6,-0.25),new THREE.Vector3(-4,2,-0.25),new THREE.Vector3(-4,2,5),new THREE.Vector3(0,0,5)];
```

```
var greenGeo2=new THREE.BufferGeometry().setFromPoints(greenPts2);
```

```
mainGroup.add(new THREE.Line(greenGeo2,wireGreenMat));
```

```
var orangePts2=[new THREE.Vector3(11,0,5),new THREE.Vector3(11,0,-5),new THREE.Vector3(0,0,-5),new THREE.Vector3(0,5,-5)];
```

```
var orangeGeo2=new THREE.BufferGeometry().setFromPoints(orangePts2);
```

```
mainGroup.add(new THREE.Line(orangeGeo2,wireOrangeMat));
```

```
var panelGeo=new THREE.BoxGeometry(20,0.15,20);
```

```
var panelMat=new  
THREE.MeshStandardMaterial({color:0x1A2840,metalness:0.4,roughness:0.3,emissive:0x0A1428,emissiveIntensity:0.1});
```

```
var panel=new THREE.Mesh(panelGeo,panelMat);
```

```
panel.position.set(0,15.2,0);
```

```
mainGroup.add(panel);
```

```
for(var i=-2;i<=2;i++){
```

```
var barGeo=new THREE.BoxGeometry(4.5,0.04,0.8);
```

```
var bar=new THREE.Mesh(barGeo,panelMat);  
bar.position.set(i*4.8,15.35,0);  
mainGroup.add(bar);  
}
```

```
var turbinePole=new THREE.Mesh(new THREE.CylinderGeometry(0.15,0.15,8,8),darkMat);  
turbinePole.position.set(12,19,0);  
mainGroup.add(turbinePole);
```

```
var turbineHub=new THREE.Mesh(new THREE.CylinderGeometry(0.3,0.3,0.3,8),bodyMat);  
turbineHub.position.set(12,23,0);  
mainGroup.add(turbineHub);
```

```
for(var b=0;b<3;b++){  
var bladeGeo=new THREE.BoxGeometry(4,0.08,0.6);  
var blade=new THREE.Mesh(bladeGeo,bodyMat);  
blade.position.set(12,23,0);  
blade.rotation.y=b*(Math.PI*2/3);  
blade.position.x+=Math.cos(b*(Math.PI*2/3))*2;  
blade.position.z+=Math.sin(b*(Math.PI*2/3))*2;  
mainGroup.add(blade);  
}
```

```
var gndGeo=new THREE.PlaneGeometry(120,120);  
var gndMat=new THREE.MeshStandardMaterial({color:0xE8E0D4,roughness:1,metalness:0});  
var ground=new THREE.Mesh(gndGeo,gndMat);  
ground.rotation.x=-Math.PI/2;  
ground.position.y=-0.01;  
ground.receiveShadow=true;  
scene.add(ground);
```

```
scene.position.set(-11,-2,-11);
```

```
scrollSteps=[
```

```
{trigger:'#step-1',pos:new THREE.Vector3(-5,35,45),look:new THREE.Vector3(0,5,0),effect:null},
```

```
{trigger:'#step-2',pos:new THREE.Vector3(-25,12,30),look:new  
THREE.Vector3(0,7.5,0),effect:'opacity'},
```

```
{trigger:'#step-3',pos:new THREE.Vector3(0,9,28),look:new THREE.Vector3(0,7.5,5),effect:null},
```

```
{trigger:'#step-4',pos:new THREE.Vector3(-4,10,15),look:new THREE.Vector3(-4,6,4),effect:'glow'},
```

```
{trigger:'#step-5',pos:new THREE.Vector3(11,15,-35),look:new THREE.Vector3(11,2.5,0),effect:null}
```

```
];
```

```
var targetPos=new THREE.Vector3(-5,35,45);
```

```
var targetLook=new THREE.Vector3(0,5,0);
```

```
var scrollActive=false;
```

```
var scrollEndTimeout=null;
```

```
gsap.registerPlugin(ScrollTrigger);
```

```
scrollSteps.forEach(function(step){
```

```
ScrollTrigger.create({
```

```
trigger:step.trigger,
```

```
start:'top center',
```

```
end:'bottom center',
```

```
scrub:0.8,
```

```
onEnter:function(){
```

```
scrollActive=true;
```

```
controls.enabled=false;
```

```
clearTimeout(scrollEndTimeout);
```

```
},
```

```
onUpdate:function(self){
```

```
targetPos.lerp(step.pos,self.progress);
```

```

targetLook.lerp(step.look,self.progress);
if(step.effect==='opacity'){
box1.material.opacity=1-(self.progress*0.85);
}else{
box1.material.opacity=Math.min(box1.material.opacity+0.02,1);
}
if(step.effect==='glow'){
coreMat.emissiveIntensity=0.4+self.progress*1.2;
coreInnerMat.emissiveIntensity=0.8+self.progress*1.5;
}else{
coreMat.emissiveIntensity=Math.max(coreMat.emissiveIntensity-0.02,0.4);
coreInnerMat.emissiveIntensity=Math.max(coreInnerMat.emissiveIntensity-0.02,0.8);
}
},
onLeave:function(){
clearTimeout(scrollEndTimeout);
scrollEndTimeout=setTimeout(function(){
scrollActive=false;
controls.enabled=true;
},700);
}
});
});

var stepBlocks=document.querySelectorAll('.step-block');
var stepObs=new IntersectionObserver(function(entries){
entries.forEach(function(e){
if(e.isIntersecting){e.target.classList.add('visible')}
});
},{threshold:0.15});
stepBlocks.forEach(function(b){stepObs.observe(b)});

```

```
var clock=new THREE.Clock();
```

```
function animate(){  
  requestAnimationFrame(animate);  
  if(!camera || !scene || !controls || !scrollSteps.length)return;  
  var dt=clock.getDelta();  
  if(scrollActive){  
    camera.position.lerp(targetPos,0.04);  
    controls.target.lerp(targetLook,0.04);  
  }  
  controls.update();  
  lidPivot.rotation.x+=(-Math.PI*0.12-lidPivot.rotation.x)*0.005;  
  coreInner.rotation.y+=0.003;  
  renderer.render(scene,camera);  
}  
animate();
```

```
window.addEventListener('resize',function(){  
  camera.aspect=window.innerWidth/window.innerHeight;  
  camera.updateProjectionMatrix();  
  renderer.setSize(window.innerWidth>window.innerHeight);  
});
```

```
var telemVals=document.querySelectorAll('.telemetry-val');  
setInterval(function(){  
  if(!telemVals.length)return;  
  telemVals[0].textContent=(4.7+Math.random()*0.3).toFixed(2)+'V';  
  telemVals[1].textContent=(11.5+Math.random()*2.5).toFixed(1)+' mA';  
  telemVals[2].textContent=(55+Math.random()*10).toFixed(1)+' mW';  
  telemVals[3].textContent=(40+Math.floor(Math.random()*5))+'%';
```

```
telemVals[4].textContent=(22.5+Math.random()*2).toFixed(1)+' C';
var bar=document.querySelector('.bg-\\[\\#4D7C0F\\]');
if(bar){bar.style.width=(60+Math.floor(Math.random()*20))+'%'}
},2000);

});
</script>
</body>
</html>
'''
```